

Correction Robot UR5

1)

	θ_i	d_i	a_i	α_i
1	θ_1	l_1	0	$\pi/2$
2	θ_2	0	$-l_2$	0
3	θ_3	0	$-l_3$	0
4	θ_4	l_4	0	$\pi/2$
5	θ_5	l_5	0	$-\pi/2$
6	θ_6	l_6	0	0

2)

$$q = \begin{bmatrix} 0 \\ -\pi/2 \\ 0 \\ -\pi/2 \\ 0 \\ \pi \end{bmatrix}$$

$$\begin{aligned}
 {}^0T_1 &= \begin{bmatrix} C_1 & 0 & S_1 & 0 \\ S_1 & 0 & -C_1 & 0 \\ 0 & 1 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} & {}^1T_2 &= \begin{bmatrix} C_2 & -S_2 & 0 & -l_2 C_2 \\ S_2 & C_2 & 0 & -l_2 S_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 {}^2T_3 &= \begin{bmatrix} C_3 & -S_3 & 0 & -l_3 C_3 \\ S_3 & C_3 & 0 & -l_3 S_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & {}^3T_4 &= \begin{bmatrix} C_4 & 0 & S_4 & 0 \\ S_4 & 0 & -C_4 & 0 \\ 0 & 1 & 0 & l_4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 {}^4T_5 &= \begin{bmatrix} C_5 & 0 & -S_5 & 0 \\ S_5 & 0 & C_5 & 0 \\ 0 & -1 & 0 & l_5 \\ 0 & 0 & 0 & 1 \end{bmatrix} & {}^5T_6 &= \begin{bmatrix} C_6 & -S_6 & 0 & 0 \\ S_6 & C_6 & 0 & 0 \\ 0 & 0 & 1 & l_6 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 {}^0R_6 &= Rot(\vec{z}, \phi) Rot(\vec{y}, \theta) Rot(\vec{z}, \psi) \\
 &= \begin{bmatrix} C_\phi & -S_\phi & 0 \\ S_\phi & C_\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} C_\theta & 0 & S_\theta \\ 0 & 1 & 0 \\ -S_\theta & 0 & C_\theta \end{bmatrix} \begin{bmatrix} C_\psi & -S_\psi & 0 \\ S_\psi & C_\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} C_\phi C_\theta C_\psi - S_\phi S_\psi & -C_\phi C_\theta S_\psi - S_\phi C_\psi & C_\phi S_\theta \\ S_\phi C_\theta C_\psi - C_\phi S_\psi & -S_\phi C_\theta S_\psi + C_\phi C_\psi & S_\phi S_\theta \\ -S_\theta C_\psi & S_\theta S_\psi & C_\theta \end{bmatrix}
 \end{aligned}$$

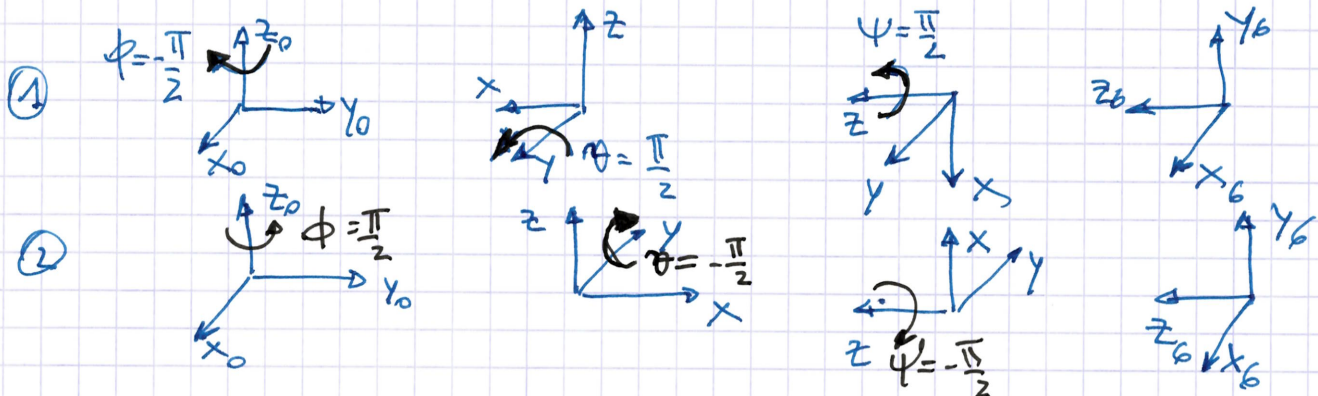
$$5) {}^0R_6 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} = \text{Rot}(\vec{x}, \pi/2).$$

$\begin{matrix} \uparrow & \uparrow & \uparrow \\ x_6 & y_6 & z_6 \end{matrix}$

$${}^0T_6 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -l_4 - l_6 \\ 0 & 1 & 0 & l_1 + l_2 + l_3 + l_5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Per identification terme à terme de 0R_6

$$\begin{aligned} \cos\theta &= 0 \longrightarrow \theta = \pm \frac{\pi}{2} \\ \sin\phi \sin\theta &= -1 \\ \sin\psi \sin\theta &= 1 \end{aligned} \quad \begin{aligned} &\theta = \frac{\pi}{2}, \psi = \frac{\pi}{2}, \phi = -\frac{\pi}{2} \quad ① \\ &\theta = -\frac{\pi}{2}, \psi = -\frac{\pi}{2}, \phi = +\frac{\pi}{2} \quad ② \end{aligned}$$



$$6) {}^0T_6 = \begin{bmatrix} {}^0R_6 & \begin{matrix} p_x \\ p_y \\ p_z \end{matrix} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} X_{\text{figure}} &= \begin{bmatrix} 0, -l_4 - l_6, l_1 + l_2 + l_3 + l_5, +\frac{\pi}{2}, -\frac{\pi}{2}, -\frac{\pi}{2} \end{bmatrix}^T \\ m &= \begin{bmatrix} -, -, -, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{\pi}{2} \end{bmatrix}^T \end{aligned}$$

$$7) {}^0T_6(q) = {}^0T_1(q_1) {}^1T_2(q_2) {}^2T_3(q_3) {}^3T_4(q_4) {}^4T_5(q_5) {}^5T_6(q_6)$$

$$\parallel$$

$${}^0T_6(x)$$

Per identification

$$\begin{cases} P_x \\ P_y \\ P_z \\ \phi \\ \psi \\ \chi \end{cases} = F(\theta_1, \dots, \theta_6) = F(q)$$

8) Pas tout à fait, car les axes 4, 5 et 6 ne sont pas tous concourants en un pt unique (centre du poignet).

$$9) {}^3T_6 = {}^3T_4 \left(\frac{\pi}{2} \right) {}^4T_5 \left(\frac{\pi}{2} \right) {}^5T_6(q) = \begin{bmatrix} 0 & 1 & 0 & -l_5 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & +l_4 + l_6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$10) {}^0T_6 = {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_6$$

$$= \begin{bmatrix} -S_{23} & C_{23} & 0 & -l_2 C_2 - (l_3 + l_5) C_{23} \\ 0 & 0 & -1 & -l_4 - l_6 \\ C_{23} & S_{23} & 0 & l_1 - l_2 S_2 - (l_3 + l_5) S_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

P_x
 P_y
 P_z

Seuls P_x et P_z sont commandables.

$$11) \begin{pmatrix} \dot{p}_x \\ \dot{p}_y \\ \dot{p}_z \end{pmatrix} = \begin{pmatrix} l_2 s_2 + (l_3 + l_5) s_{23} & (l_3 + l_5) s_{23} \\ 0 & 0 \\ -l_2 c_2 - (l_3 + l_5) c_{23} & -(l_3 + l_5) c_{23} \end{pmatrix} \begin{pmatrix} \dot{\theta}_2 \\ \dot{\theta}_3 \end{pmatrix}$$

$\uparrow$
 ${}^0 \vec{z}_2 \times {}^0 \vec{O}_2 \vec{O}_6$

$\uparrow$
 ${}^0 \vec{z}_3 \times {}^0 \vec{O}_3 \vec{O}_6$

$$12) \begin{pmatrix} \dot{p}_x \\ \dot{p}_z \end{pmatrix} = J_r \begin{pmatrix} \dot{\theta}_2 \\ \dot{\theta}_3 \end{pmatrix}$$

$$\det(J_r) = l_2(l_3 + l_5) s_3.$$

$$\det(J_r) = 0 \text{ si } \theta_3 = 0 \text{ ou } \pi.$$

Singularité des limites
de l'espace atteignable.

13) voir cours.